

Correlation and Covariance Assignment

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Answer: Covariance is a statistical measure that indicates the direction of the linear relationship between two variables. It measures how much two variables change together — if one variable increase and the other also increases, the covariance is positive; if one increases while the other decreases, it is negative.

Formula:

$$\text{Cov}(X, Y) = \Sigma [(X_i - \bar{X})(Y_i - \bar{Y})] / n$$

Correlation is a standardized version of covariance. It measures both the direction and strength of the linear relationship between two variables, and always lies between -1 and $+1$.

Formula:

$$r = \text{Cov}(X, Y) / (\sigma_x \times \sigma_y)$$

Differences between Covariance and Correlation:

Feature	Covariance	Correlation
Scale	Unbounded ($-\infty$ to $+\infty$)	Fixed between -1 and $+1$
Unit	Has units (product of both variables' units)	Unitless (dimensionless)
Interpretation	Direction only	Direction + Strength
Comparability	Cannot compare across datasets	Can compare across datasets
Effect of scaling	Changes if variables are scaled	Unaffected by scaling

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Answer: The sign of covariance reveals the direction of the relationship between two variables X and Y:

1.) Positive Covariance — $\text{Cov}(X, Y) > 0$

Indicates that both variables move in the same direction. When X increases, Y tends to increase as well, and when X decreases, Y tends to decrease.

Example: Hours studied and marks obtained — as study hours increase, marks tend to increase too.

2. Negative Covariance — $\text{Cov}(X, Y) < 0$

Indicates that both variables move in opposite directions. When X increases, Y tends to decrease, and vice versa.

Example: Number of absences and marks obtained — as absences increase, marks tend to decrease.

3. Zero Covariance — $\text{Cov}(X, Y) = 0$

Indicates no linear relationship between the two variables. Changes in X do not correspond to any consistent change in Y.

Example: Shoe size and intelligence — there is no consistent relationship between them.

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Answer: While covariance is a useful measure to determine the direction of a relationship between two variables, it has several significant limitations that reduce its practical usefulness. Which are mentioned as below:

Limitations of Covariance:

1. Scale Dependency: Covariance is highly sensitive to the units of measurement. If the units of either variable are changed (e.g., converting kilograms to grams), the covariance value changes drastically, even though the actual relationship between the variables remains the same.

2. Lack of Standardization: The value of covariance ranges from $-\infty$ to $+\infty$. There is no fixed boundary, which makes it impossible to judge whether a covariance value is large or small in absolute terms.

3. No Measure of Strength: Covariance only reveals the direction of a relationship (positive or negative) but gives no information about the strength of that relationship. A covariance of 500 does not necessarily indicate a stronger relationship than a covariance of 5.

4. Not Comparable Across Datasets: Because covariance depends on units and scale, it cannot be used to compare relationships across different datasets or different pairs of variables.

5. Sensitive to Outliers: Covariance is significantly affected by extreme values (outliers) in the data, which can distort the result and give a misleading picture of the relationship.

6. Only Captures Linear Relationships: Covariance can only detect linear relationships between variables. If two variables have a strong non-linear relationship, covariance may be close to zero, completely missing the pattern.

Why Correlation is Preferred:

Correlation overcomes all the major limitations of covariance by standardizing the measure:

$$r = \text{Cov}(X, Y) / (\sigma_x \times \sigma_y)$$

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Answer: Both Pearson's and Spearman's correlation coefficients measure the relationship between two variables, but they differ fundamentally in their assumptions, data requirements, and methodology.

Pearson's Correlation Coefficient (r)

Pearson's correlation measures the linear relationship between two continuous variables. It works directly with the actual values of the data and assumes that the relationship between variables is a straight line.

Formula:

$$r = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sqrt{[\sum(X_i - \bar{X})^2 \times \sum(Y_i - \bar{Y})^2]}}$$

Assumptions of Pearson's:

- Both variables must be continuous
- Data must be normally distributed
- The relationship must be linear
- No significant outliers should be present

Spearman's Rank Correlation Coefficient (r_s)

Spearman's correlation measures the monotonic relationship between two variables — meaning it checks whether the variables tend to move in the same direction, but not necessarily at a constant rate. It works by first converting data into ranks and then applying correlation on those ranks.

Formula:

$$r_s = 1 - [6 \sum d^2] / [n(n^2 - 1)]$$

Where **d** = difference between ranks of each pair, and **n** = number of observations.

Assumptions of Spearman's:

- Data can be ordinal, interval, or ratio
- Does not require normal distribution
- Captures monotonic (not just linear) relationships
- More robust to outliers

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Answer: Given: Correlation coefficient **r = 0.85** between variables X and Y

Interpretation of r = 0.85:

A correlation coefficient of **0.85** indicates a **strong positive linear relationship** between variables X and Y. This means:

- As variable X increases, variable Y also tends to increase
- The relationship is **strong** because the value is close to +1
- Approximately **72.25%** of the variation in Y can be explained by X

Calculated as:

$r^2 = (0.85)^2 = 0.7225 \rightarrow 72.25\%$ this is called the Coefficient of Determination

Contextual Example: Suppose X = Hours Studied and Y = Marks Obtained by students. An $r = 0.85$ tells us that students who study more hours tend to score significantly higher marks, and this pattern is strong and consistent across the dataset.

Strength Classification of $r = 0.85$:

Can We Infer Causation from $r = 0.85$?

No. Correlation does not imply causation.

Even though $r = 0.85$ shows a strong relationship between X and Y , it does **not** prove that X *causes* Y . There are three key reasons for this:

1. Direction of Causation is Unknown The correlation does not tell us which variable is influencing the other. It is equally possible that Y causes X , or that neither causes the other directly.

2. Presence of Confounding Variables A third hidden variable (called a **lurking or confounding variable**) may be causing both X and Y to move together, creating the illusion of a direct relationship.

Example: Ice cream sales and drowning incidents are strongly correlated — but ice cream does not cause drowning. The hidden variable is **summer/hot weather**, which increases both independently.

3. Spurious Correlation Two completely unrelated variables can show a high correlation purely by coincidence, especially in small datasets. This is called a **spurious correlation**.

Example: The number of movies Nicolas Cage appeared in per year has been found to correlate with swimming pool drownings — clearly not a causal relationship.

Q6. Using the dataset below, calculate the covariance between X and Y .

X	2	4	6	8
Y	3	7	5	10

Answer: Formula = $\text{Cov}(X, Y) = \Sigma [(X_i - \bar{X})(Y_i - \bar{Y})] / n$

Step 1: Calculate Mean of X and Y

$$\bar{X} = (2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$$

$$\bar{Y} = (3 + 7 + 5 + 10) / 4 = 25 / 4 = 6.25$$

Step 2: Calculate $(X_i - \bar{X})$ and $(Y_i - \bar{Y})$ for each data point

X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$
2	3	$2 - 5 = -3$	$3 - 6.25 = -3.25$	$(-3)(-3.25) = 9.75$
4	7	$4 - 5 = -1$	$7 - 6.25 = +0.75$	$(-1)(0.75) = -0.75$
6	5	$6 - 5 = +1$	$5 - 6.25 = -1.25$	$(1)(-1.25) = -1.25$
8	10	$8 - 5 = +3$	$10 - 6.25 = +3.75$	$(3)(3.75) = 11.25$

Step 3: Sum of $(X_i - \bar{X})(Y_i - \bar{Y})$

$$\Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] = 9.75 + (-0.75) + (-1.25) + 11.25 = 19.00$$

Step 4: Divide by n

$$\text{Cov}(X, Y) = 19.00 / 4 = 4.75$$

Covariance between X and Y = 4.75

Q7. Compute the Pearson correlation coefficient between variables A and B:

A	10	20	30	40	50
B	8	14	18	24	28

Answer: Formula: $r = \text{Cov}(A, B) / (\sigma_A \times \sigma_B)$

Step 1: Calculate Mean of A and B

$$\bar{A} = (10 + 20 + 30 + 40 + 50) / 5 = 150 / 5 = 30$$

$$\bar{B} = (8 + 14 + 18 + 24 + 28) / 5 = 92 / 5 = 18.4$$

Step 2: Build the Full Calculation Table

A_i	B_i	$A_i - \bar{A}$	$B_i - \bar{B}$	$(A_i - \bar{A})(B_i - \bar{B})$	$(A_i - \bar{A})^2$	$(B_i - \bar{B})^2$
10	8	-20	-10.4	208.0	400	108.16
20	14	-10	-4.4	44.0	100	19.36
30	18	0	-0.4	0.0	0	0.16
40	24	+10	+5.6	56.0	100	31.36
50	28	+20	+9.6	192.0	400	92.16

Step 3: Calculate the Sums

$$\Sigma[(A_i - \bar{A})(B_i - \bar{B})] = 208.0 + 44.0 + 0.0 + 56.0 + 192.0 = 500$$

$$\Sigma(A_i - \bar{A})^2 = 400 + 100 + 0 + 100 + 400 = 1000$$

$$\Sigma(B_i - \bar{B})^2 = 108.16 + 19.36 + 0.16 + 31.36 + 92.16 = 251.20$$

Step 4: Apply the Formula

$$r = 500 / \sqrt{(1000 \times 251.20)}$$

$$r = 500 / \sqrt{(251200)}$$

$$r = 500 / 501.20$$

$$r = 0.9976 \approx 0.998$$

Pearson Correlation Coefficient $r = 0.998$

Q8. The following table shows heights (in cm) and weights (in kg) of 5 students.^[1]^[SEP] Find the correlation coefficient between Height and Weight.

Height	150	160	165	170	180
Weight	50	55	58	62	70

Answer: Formula: $r = \frac{\sum[(H_i - \bar{H})(W_i - \bar{W})]}{\sqrt{[\sum(H_i - \bar{H})^2 \times \sum(W_i - \bar{W})^2]}}$

Step 1: Calculate Mean of Height and Weight

$$\bar{H} = (150 + 160 + 165 + 170 + 180) / 5 = 825 / 5 = 165$$

$$\bar{W} = (50 + 55 + 58 + 62 + 70) / 5 = 295 / 5 = 59$$

Step 2: Build the Full Calculation Table

H_i	W_i	$H_i - \bar{H}$	$W_i - \bar{W}$	$(H_i - \bar{H})(W_i - \bar{W})$	$(H_i - \bar{H})^2$	$(W_i - \bar{W})^2$
150	50	-15	-9	135	225	81
160	55	-5	-4	20	25	16
165	58	0	-1	0	0	1
170	62	+5	+3	15	25	9
180	70	+15	+11	165	225	121

Step 3: Calculate the Sums

$$\sum[(H_i - \bar{H})(W_i - \bar{W})] = 135 + 20 + 0 + 15 + 165 = 335$$

$$\sum(H_i - \bar{H})^2 = 225 + 25 + 0 + 25 + 225 = 500$$

$$\sum(W_i - \bar{W})^2 = 81 + 16 + 1 + 9 + 121 = 228$$

Step 4: Apply the Formula

$$r = 335 / \sqrt{(500 \times 228)}$$

$$r = 335 / \sqrt{(114000)}$$

$$r = 335 / 337.64$$

$$r = 0.9922 \approx 0.992$$

Pearson Correlation Coefficient $r = 0.992$

Q9. Given the dataset below, determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Answer:

Mean-Based Reasoning (without full calculation):

$$\bar{X} = (1+2+3+4+5) / 5 = 3$$

$$\bar{Y} = (15+12+9+7+3) / 5 = 46/5 = 9.2$$

X_i Position vs $\bar{X}$ Y_i Position vs $\bar{Y}$ Same direction?

1 Below average 15 Above average ✗ Opposite

2 Below average 12 Above average ✗ Opposite

X_i Position vs $\bar{X}$ Y_i Position vs $\bar{Y}$ Same direction?

3 Equal to average 9 Below average — Neutral

4 Above average 7 Below average ✗ Opposite

5 Above average 3 Below average ✗ Opposite

In every case, when X is below its mean, Y is above its mean — and when X is above its mean, Y is below its mean. This is the textbook signature of a negative correlation.

Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Answer:

Step 1: Calculate Means

$$\bar{A} = (8 + 10 + 12 + 9 + 11) / 5 = 50 / 5 = 10$$

$$\bar{B} = (6 + 9 + 11 + 8 + 10) / 5 = 44 / 5 = 8.8$$

Step 2: Build the Full Calculation Table

Year A_i B_i $A_i - \bar{A}$ $B_i - \bar{B}$ $(A_i - \bar{A})(B_i - \bar{B})$ $(A_i - \bar{A})^2$ $(B_i - \bar{B})^2$

1 8 6 -2 -2.8 5.6 4 7.84

2 10 9 0 +0.2 0.0 0 0.04

3 12 11 +2 +2.2 4.4 4 4.84

4 9 8 -1 -0.8 0.8 1 0.64

5 11 10 +1 +1.2 1.2 1 1.44

Step 3: Calculate the Sums

$$\Sigma[(A_i - \bar{A})(B_i - \bar{B})] = 5.6 + 0.0 + 4.4 + 0.8 + 1.2 = 12.0$$

$$\Sigma(A_i - \bar{A})^2 = 4 + 0 + 4 + 1 + 1 = 10$$

$$\Sigma(B_i - \bar{B})^2 = 7.84 + 0.04 + 4.84 + 0.64 + 1.44 = 14.80$$

Step 4: Calculate Covariance

$$\text{Cov}(A, B) = \Sigma[(A_i - \bar{A})(B_i - \bar{B})] / n$$

$$\text{Cov}(A, B) = 12.0 / 5 = 2.4$$

Step 5: Calculate Correlation Coefficient

$$r = \Sigma[(A_i - \bar{A})(B_i - \bar{B})] / \sqrt{[\Sigma(A_i - \bar{A})^2 \times \Sigma(B_i - \bar{B})^2]}$$

$$r = 12.0 / \sqrt{(10 \times 14.80)}$$

$$r = 12.0 / \sqrt{148}$$

$$r = 12.0 / 12.166$$

$$r = 0.9864 \approx 0.986$$